Automation and the Commodification of Labor

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Companies buy human time just as they buy steel, paper, or any other commodity. There are, of course, quite a few regulations on *how* companies can buy it, but the principle is the same regardless. This means that, in order to survive in this system, humans have no choice but to sell their time. For the vast majority of humanity, this has been a necessary trade; in order to consume, one must produce. Applying capitalism to labor thus gives a convenient way to disseminate a finite pool of resources across the populace: those who produce the most are allowed to consume the most—in theory, at least.

But notice what this quietly assumes: that human labor is the only means capable of producing what we need. For all of recorded history that assumption has held, and so we built our entire moral and economic order on top of it. It is now failing.

As we will come to see, this system—the commodification of labor—is on the precipice of collapse, and the 21st century is liable to deal the final blow. Controlling whether the collapse manifests as catastrophe or liberation, I argue, will be the defining issue of the 21st century.

In 1996, Kodak was thriving; it boasted annual revenue of $16 billion and was a staple in every household—right up until digital photography debuted. Its stock nosedived, earnings dropped, and, despite its efforts to shift to digital, it eventually filed for bankruptcy [16]. Nowadays, except for a few hobbyists like myself, nobody uses film anymore. It was replaced by a cheaper, more convenient alternative. Few feel bad for Kodak; after all, this is the circle of life in the free market. Industry disruptors challenge established names until they themselves are challenged. Take music, for instance: tapes killed vinyl, CDs killed tapes, iPods killed CDs, and streaming killed iPods.

Now, what about humans—the most versatile commodity a company can buy? What happens when, like film, there is a cheaper, more convenient alternative to humans?

During the first Industrial Revolution, this question was prominent. The need for physical human labor plummeted as machines were able to work longer and harder while costing less. As a result, simple, repetitive tasks like plowing and weaving were automated away entirely [2]. As the number of automated tasks grew, humans had to shift away from using their muscles, towards leveraging their minds. Now, in today's developed world, we are effectively never employed without our intelligence. This is what continues to set us apart from machines; even seemingly simple tasks, such as delivering packages or plumbing a house, require enough inference and fine motor skill to stave off automation—at least for a while.

Today, with the advent of artificial intelligence and the alarmingly rapid rate at which it is improving, we must wonder: like our muscles, are our minds liable to be replaced?

We humans have shown ourselves to be versatile; we have always been able to outperform machines in some way. But if AI continues to grow more intelligent and robots continue to learn new skills, there will eventually—maybe not in our lifetimes, but eventually—come a day when, like Kodak film, humans have become an obsolete commodity.

Artificial intelligence has, for years, been notoriously fear-inducing. Well before the 2022 release of ChatGPT, prominent figures like Stephen Hawking [4] and Bill Gates [7] cautioned against an AI-driven doomsday.

Nick Bostrom, a prominent AI philosopher, is famous for his "Paperclip Maximizer" story, in which an AI told to manufacture paperclips eventually converts all available earth and space into paperclips or paperclip factories [3]. Bostrom is a notable philosopher, and he might be correct, but his doomsday stories border on outlandish. To think that an AI could be so smart yet do something so ridiculous is hard to believe.

There are some more legitimate concerns: cyberattacks, mass surveillance, etc. Seeing hyper-intelligent AI in this more realistic sense still allows us to arrive at a tangential conclusion: AI, particularly AGI[1], might pose an existential risk to humanity. Hence, like all existential threats, we should indeed be conservative and careful when developing AGI—if at all.

Importantly, however, the artificial intelligence that exists today is nowhere near being hyper-intelligent. It may also be the case that hyper-intelligence is not possible to achieve. Even then, moderate-intelligence automation still has the potential to eliminate the majority of jobs that exist today. Thus, under our current economic system, a working-class doomsday already seems inevitable.

The idea is relatively simple: company executives will stop hiring people once a cheaper autonomous alternative exists. These executives would then amass an unbelievable amount of capital while the workforce is forced into poverty. Then, without any route for upward mobility, the proletariat is left permanently poor and powerless while a select few families control the nation's capital. In effect, we revert to a peasant-lord society.

In 2021, OpenAI CEO Sam Altman published the essay "Moore's Law for Everything" [1]. In it, he (quixotically) dreams of a future of infinite abundance. Yet he himself arrives at this same problem, and—surprisingly for a billionaire—his solution is a form of aggressive socialism:

> *"We should therefore focus on taxing capital rather than labor, and we should use these taxes as an opportunity to directly distribute ownership and wealth to citizens. In other words, the best way to improve capitalism is to enable everyone to benefit from it directly as an equity owner."* [1]

Viewed through the lens of a common-ownership society, automation is suddenly not a problem. Instead it's quite the opposite: the proletariat is able to enjoy the abundance that automation creates instead of being ruined by it.

[1] Artificial General Intelligence

This is a critical point: under capitalism, any automation—even if it doesn't cause a Bostrom doomsday—will directly hurt the working class. Under shared ownership, the inverse is true: the resulting increase in economic productivity is mirrored as an increase in the quality of life for the people. Karl Marx describes this fundamental failing of capitalism in *Grundrisse*:

> *"Capital itself is the moving contradiction, [in] that it presses to reduce labour time to a minimum, while it posits labour time, on the other side, as sole measure and source of wealth."* [11, p. 706]

This contradiction, coupled with the catalyst of automation, drives capitalism straight into a brick wall. The result will be a devastatingly large wealth and power gap, the likes of which the developed world has yet to see. Bill Gates, also aware of the issue, comments on it:

> *"AI will either be the greatest equalizer ever invented, or the worst source of injustice."* [8]

Either we allow capitalism to draw a permanent line between classes, or we develop methods to equitably disperse capital and ownership to the people.

It is important to note that many doubt this will ever happen—that a permanent underclass will develop. The "lump of labor fallacy" is the term economists use to describe this exact line of thought [18]. The Federal Reserve Bank of St. Louis explains just this:

> *"In the year 1900, 41 percent of the U.S. workforce worked in agriculture. After a century of technological change in that industry, the number stood at 2 percent (in the year 2000). This transformation changed the work of farmers in dramatic ways, but it did not reduce total employment in the United States. As it happens, the mechanization of agriculture in the early twentieth century made possible the large increase in employment in new industry and factory jobs."* [18]

True, so far humans have been able to find other jobs relatively easily when one gets automated away. The logic follows the idea that there is not a fixed amount of work to do; therefore, when something is automated away, something new is created, and people are again employed.

But this theory is hand-wavy and grounded primarily in observation of the past. Simply because humans have historically been able to *adapt* and outperform machines at some other task does not guarantee that the trend will continue. Already, as established above, simple muscle-driven labor, like manual farming and mining, has been automated away almost entirely in the U.S., except in cases that require fine motor skills, such as stocking shelves or working in construction. Most people transitioned into intellectual fields like software engineering. But with advances in AI, these "intellectual" jobs are suddenly liable to be automated away as well.

The economist citing the lump of labor fallacy would argue that history repeats itself and these people will simply go on to get other jobs—often better ones too. After all, this has been the established pattern. If this is indeed the case, great, but eventually there *is* an automation endgame in which humans lose the labor battle. We are only winning

today because we were able to pivot toward intellectual work. It is now the robots' turn to follow suit. Some jobs—law, medicine, politics—will likely never disappear entirely, but it is improbable, and frankly ridiculous, to assume the rest of the populace could absorb into one of these roles; we simply do not need 8 billion doctors, lawyers, and politicians.

A more scholarly counter may pull comparative advantage into the conversation. Economic theory dictates that, even if machines outperform humans at everything, humans retain a relative advantage somewhere, and so there will always be work for them. This is true and almost entirely beside the point. Comparative advantage guarantees a person a *task*; it says nothing whatsoever about what that task pays. Horses retained a comparative advantage in certain jobs well after the internal combustion engine arrived, and the horse population collapsed anyway, because the market-clearing wage fell below the cost of feeding and stabling them [10]. Nothing in the theorem places the human wage floor above subsistence.

Some people argue, very reasonably, that we should play it safe and simply stop the development of autonomous technologies, especially artificial intelligence. While regulation is certainly a necessity, this train of thought is flawed for two reasons. First, I will argue that efforts to pause the development of these technologies are futile. Then I'll show that, even if the technologies *could* be stopped, it is not in our interest as a nation to do so.

On the first matter—the futility of halting these technologies—consider the Luddites. This movement took place in the early 19th century, when the automation of the textile industry left weavers without work. These workers coalesced and rebelled violently against their employers for replacing them, but their efforts were futile [17].

The term "Luddite" has grown to encapsulate a general movement against technology. Neo-Luddism, the contemporary adaptation, is diverse and leaderless, but its members share a disdain for modern technology and globalization. Their grievances are generally derived from capitalist exploitation and environmental concerns. Because of this, the Neo-Luddites fill a wide spectrum, ranging from radical environmentalists to off-the-grid dwellers. In almost all cases, they are anxious for radical, often violent change.

To quickly put this to rest: Luddism is not the answer. Under democratic rule, a complete rejection of technology is extraordinarily unpopular within the electorate, so any effective effort would need to bypass democracy. This could take myriad forms. Eco-fascism—a form of fascism that sacrifices humans and technology for the sake of the planet—is one. Another is accelerationist anarchism: the belief that the faster modern society falls, the faster humans can return to a just, pre-tech way of life. The Unabomber, Ted Kaczynski, is a famous Neo-Luddite anarchist who waged a 17-year mail-bombing campaign in an attempt to accelerate society's collapse [5, 9].

However, not all anti-tech movements are as radical as Luddism. Many contemporary movements, such as the STOP AI organization, propose a much more moderate solution: simply prevent companies from continuing to develop AI [13]. A similar campaign, PAUSE AI, advocates pausing AI development until it seems safe to proceed [12].

Both organizations share similar concerns—mostly about job loss—but also cite Bostrom-style doomsday scenarios as a reason to hold off. These movements stand a fair shot at winning democratic support. They are certainly underdogs, and without a strong financial

backbone their chances of persuading the American legislature are next-to-none—but not impossible. Still, as we will see, this approach is not merely unlikely to succeed; it has the potential to be radically dangerous.

Let us see AI in the most negative light: a technology capable of causing doomsday with zero upside. Any isolated nation with sane leadership would immediately halt its development. However, the United States is far from an isolated nation, and one of our biggest adversaries, China, is in an AI arms race with us.

During the nuclear arms race of the Cold War, it would have seemed, and been, ridiculous to propose that the United States should simply halt the development of its weapons. To state the obvious, that would have allowed our adversary, the Soviets, to out-arm and subsequently outpower us, and the U.S. would become deferential to the Soviets.

This is not to defend U.S. imperialism. Current world affairs clearly show how miserable it is, as a nation and a people, to be forced to submit to a global superpower—in this case, the United States. As Americans, we are exceedingly fortunate not to need to exercise any foreign deference and, most importantly, we hold the incredible prerogative to vote away and limit the abuse—even if, for whatever reason, we are yet to. Imperialism is evidently terrible, and Americans should strive to release their grasp on the rest of the world through controlled means. That, however, does not mean forfeiting that power to an equally or more dubious foreign adversary like the Soviets or the Chinese.

So if America halts domestic AI development without its technological adversaries agreeing to follow suit, we immediately put ourselves at risk. One side does not simply lay down its arms. Instead, negotiation to regulate AI—ensuring a doomsday catastrophe doesn't kill us both, and that neither side out-powers the other—is necessary, just as it was to eventually put an end to the Cold War's arms race [15].

In the highly unlikely scenario that the U.S. and China both agree to halt the development of AI, one major moral issue remains. To me, this is the most fundamental. The way things are today, for so many people, is not good enough. In fact, I believe it is immoral to artificially force people to linger in this era of global injustice any longer than necessary.

With the exception of the few who are born or marry rich, anyone able to work is forced to work for their entire life up until retirement. Just about everyone has seen firsthand how much dread comes from the necessity of employment. Personally, I think of all the colorful people I knew who, when eventually thrust into the workforce, dropped their hobbies, friendships, and happiness. Worse, we continue to force humans into dangerous jobs in warehouses and on farms; in fact, a worker died every 104 minutes from a work-related injury in 2024 [14]. We have forced our entire populace to live against their deepest wishes, so it is no wonder that 2 in 10 Americans are depressed [6].

Work like this is ethical if it's necessary, but it might not always be. And to think that it *could* be unnecessary in the near future should have people across the world jumping for joy. If you could release the artists, musicians, and authors from their uninspiring jobs, wouldn't you? Imagine the renaissance that would occur if we were free to live our lives to the fullest.

To pause life in this reality, to prevent the social and artistic boom that the future may hold, simply because the United States has clung to a faulty economic system, can be described only as absurdity—particularly when the citizens hold the power to change it.

The only ethical thing to do, I argue, is to fight for progressive change—any means of redistributing capital, wealth, and ownership to the working people—while simultaneously fighting for *regulated* automation, not the absence of it. It is absurd and glaringly immoral to keep forcing all able people to enlist themselves, against their wishes, in a company until retirement when a viable alternative exists.

The automation endgame threatens humans as sellers of labor; unless there is a Bostrom doomsday, it does not itself threaten humans. The wealth that machines produce does not evaporate—it accrues to whoever owns them, and who owns them, how they are taxed, and where the proceeds go are not facts of nature but decisions, made by representatives, subject to elections. It may appear that the American legislature is incorrigible and helpless on the matter, but history repeatedly shows that, when power injustice grows obscene, people coalesce—and when people coalesce under democracy, they win.

Automation is the first force in human history powerful enough to make ending obligatory toil a practical question, rather than a utopian fantasy—and that should be cause for hope, not dread. What remains is not to stop the machines, but to decide who they serve—and that decision, unlike the technology itself, is entirely ours to make.

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